Heat and material balance

Flash drum D-400 with feed preheater E-400 · design, maximum rate and turndown cases

Document no. FD-V1-PE-HMB-001

The heat and material balance fixes, for every stream of the PFD FD-400-PFD-001 and the P&ID FD-400-PID-001, the flow, composition, temperature, pressure and phase, and the duty of E-400. Every other discipline designs from it.

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Holds and engineering queries

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Note. Numbers come from the simulator model (design case and steady model runs) and the stated assumptions; generated for teaching, not for construction. Engineering queries raised by this document are listed in it and in FD-V1-PM-DOC-001.

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DOCUMENT NO.	REV	STATUS	ORIGINATOR	DATE
FD-V1-PE-HMB-001	0	Issued for teaching	Apinan Soottitantawat	28-OCT-26

1 · Basis and cases

Feed	16 m³/h (10,208 kg/h), 35 wt% n-butane (44.4 mol%), 10 °C; supply 650 kPa g at no flow (P-400 shut-off), 600 kPa g at design
Products	vapour ≥ 75 wt% butane (AI-420), to the fuel-gas / vapour header 30 kPa g; liquid ≤ 28 wt% butane (AI-410), to the downstream receiver 40 kPa g
Hot oil	180 °C supply, header ΔP 150 kPa, c_p 2.3 kJ/kg K, 10 kg/s (36 t/h) through TCV-401 fully open; header pressures 500 / 350 kPa g (assumed)
Nitrogen	700 kPa g header (P&ID), blanket 30 kPa g
Flare	closed flare header, superimposed back-pressure 10 kPa g (assumed)
Ambient	28 °C mean (model K.Tamb); minimum 15 °C (assumed, tropical site)
Hot-oil properties	hot oil (assumed typical mineral heat-transfer oil): ρ 800 kg/m³, μ 1.0 mPa s at 120–180 °C, c_p 2.3 kJ/kg K (model K.cph)
Model	ideal Raoult VLE, Antoine fitted to saturation data, Watson latent heats, adiabatic flash; molar masses 58.12 / 86.18 kg/kmol

Case	Feed m³/h	How it was obtained	Why this case
Design case	16.0	createSim() + warmStart(600 s) + 7200 s at the design set points, noise off, trips off (converged; = the page snapDesign)	The simulator design point (legacy booklet): feed 16 m³/h, TIC-401 74 °C, PIC-420 102.7 kPa g, LIC-400 50 %.
Maximum rate	20.0	FIC-400 SP 20 m³/h, all loops in AUTO, trips off, 7200 s of plant time from design	Top of the operating window with the present alarms: FIC-400 HI alarm 20 m³/h (Operating windows tab: 9.3–20.0 m³/h with alarms; equipment and quality 6.0–24.8 m³/h).
Turndown	10.0	FIC-400 SP 10 m³/h, all loops in AUTO, trips off, 7200 s of plant time from design	Bottom of the operating window with the present alarms rounded up: FIC-410 LO alarm 8 m³/h is reached at ≈ 9.3 m³/h feed (Operating windows tab).

Design = the simulator design snapshot; maximum rate and turndown = steady runs of the same model with the trips disabled (FIC-400 set point moved, every other set point at design). The model treats the bottoms as liquid down to the receiver: stream 5A is a flash of stream 5 at the LCV-400 outlet pressure, computed with the model thermodynamics (adFlash).

2 · Stream table, design case (16 m³/h)

Stream	2	3	3A	4	5	5A	H1	H2	H3	H4	H5	N1	F1
Description	Feed to E-400 (P-400 discharge, battery limit)	Hot feed to FCV-400 (liquid under pressure)	Let-down outlet, flashing two-phase to D-400	Vapour to the fuel-gas header	D-400 liquid to LCV-400 (bubble point)	Bottoms after LCV-400 to the downstream unit (flashing)	Hot-oil supply to TCV-401 (header, total)	Hot oil to E-400 shell	Hot oil from E-400 shell	Hot-oil bypass (3-way valve)	Hot-oil return to header (mixed)	Nitrogen blanket (PCV-430)	PSV-400 relief to flare
From → to	P-400 (upstream unit) → E-400 tubes	E-400 tubes → FCV-400	FCV-400 / HV-401 → D-400 inlet N1	D-400 top (PCV-420) → Vapour header 30 kPa g	D-400 bottom N3 → LCV-400	LCV-400 / HV-411 → Downstream receiver 40 kPa g	Hot-oil supply header → TCV-401 (3-way)	TCV-401 → E-400 shell	E-400 shell → Return junction	TCV-401 → Return junction	Return junction → Hot-oil return header	N2 header 700 kPa g → D-400 top	D-400 (PSV-400) → Flare header
Phase	L	L	L+V	V	L	L+V	HO	HO	HO	HO	HO	N2	V
Vapour mass fraction	0.000	0.000	0.170	1.000	0.000	0.053	—	—	—	—	—	—	—
Temperature °C	10.0	74.0	49.3	49.2	49.2	41.4	180.0	180.0	120.1	180.0	161.7	28.0	49.2
Pressure kPa g	600.0	550.0	102.7	102.7	133.6	50.0	500.0	480.0	370.0	370.0	350.0	102.7	102.7
Mass flow kg/h	10,208	10,208	10,208	1,714	8,495	8,495	36,000	10,980	10,980	25,020	36,000	0	0
Molar flow kmol/h	138.47	138.47	138.47	27.26	111.21	111.21	—	—	—	—	—	0.00	0.00
Volume m³/h actual	16.0	17.9	376.4	358.1	14.0	135.3	45.0	13.7	13.7	31.3	45.0	0.0	—
Vapour Nm³/h	—	—	—	611	—	—	—	—	—	—	—	—	—
Density kg/m³	638.01	571.81	27.12	4.79	605.41	62.77	800.00	800.00	800.00	800.00	800.00	2.28	—
Butane wt%	35.00	35.00	35.00	76.82	26.56	26.56	—	—	—	—	—	—	—
Molar mass	73.72	73.72	73.72	62.86	76.38	76.38	—	—	—	—	—	28.01	62.86
Instruments	FT-400, TI-400	TT-401	(TI-410 in the drum)	FI-420, AI-420	FT-410, AI-410, TI-410	—	TI-402, PDI-406	FT-405	TI-403	—	—	AI-400	—
PFD stream	2	3	3	4	5	5	—	—	—	—	—	—	—

Valves: FCV-400 73.5 %, TCV-401 30.5 %, LCV-400 63.4 %, PCV-420 38.7 %. Analysers: AI-420 76.82 wt% (spec ≥ 75), AI-410 26.56 wt% (spec ≤ 28), TI-410 49.2 °C. Pressures: P-400 discharge 600.0, E-400 outlet 560.0, FCV-400 inlet 550.0 kPa g; hot-feed bubble pressure 359.1 kPa g (margin 190.9 kPa).

2 · Stream table, maximum rate (20 m³/h)

Stream	2	3	3A	4	5	5A	H1	H2	H3	H4	H5	N1	F1
Description	Feed to E-400 (P-400 discharge, battery limit)	Hot feed to FCV-400 (liquid under pressure)	Let-down outlet, flashing two-phase to D-400	Vapour to the fuel-gas header	D-400 liquid to LCV-400 (bubble point)	Bottoms after LCV-400 to the downstream unit (flashing)	Hot-oil supply to TCV-401 (header, total)	Hot oil to E-400 shell	Hot oil from E-400 shell	Hot-oil bypass (3-way valve)	Hot-oil return to header (mixed)	Nitrogen blanket (PCV-430)	PSV-400 relief to flare
From → to	P-400 (upstream unit) → E-400 tubes	E-400 tubes → FCV-400	FCV-400 / HV-401 → D-400 inlet N1	D-400 top (PCV-420) → Vapour header 30 kPa g	D-400 bottom N3 → LCV-400	LCV-400 / HV-411 → Downstream receiver 40 kPa g	Hot-oil supply header → TCV-401 (3-way)	TCV-401 → E-400 shell	E-400 shell → Return junction	TCV-401 → Return junction	Return junction → Hot-oil return header	N2 header 700 kPa g → D-400 top	D-400 (PSV-400) → Flare header
Phase	L	L	L+V	V	L	L+V	HO	HO	HO	HO	HO	N2	V
Vapour mass fraction	0.000	0.000	0.170	1.000	0.000	0.046	—	—	—	—	—	—	—
Temperature °C	10.0	74.0	49.3	49.2	49.2	42.3	180.0	180.0	124.0	180.0	157.2	28.0	49.2
Pressure kPa g	571.9	493.8	102.7	102.7	133.6	55.7	500.0	480.0	370.0	370.0	350.0	102.7	102.7
Mass flow kg/h	12,760	12,760	12,760	2,147	10,613	10,613	36,000	14,692	14,692	21,308	36,000	0	0
Molar flow kmol/h	173.08	173.08	173.08	34.16	138.93	138.93	—	—	—	—	—	0.00	0.00
Volume m³/h actual	20.0	22.3	470.5	448.7	17.5	147.1	45.0	18.4	18.4	26.6	45.0	0.0	—
Vapour Nm³/h	—	—	—	766	—	—	—	—	—	—	—	—	—
Density kg/m³	638.01	571.81	27.12	4.79	605.41	72.17	800.00	800.00	800.00	800.00	800.00	2.28	—
Butane wt%	35.00	35.00	35.00	76.82	26.54	26.54	—	—	—	—	—	—	—
Molar mass	73.72	73.72	73.72	62.86	76.39	76.39	—	—	—	—	—	28.01	62.86
Instruments	FT-400, TI-400	TT-401	(TI-410 in the drum)	FI-420, AI-420	FT-410, AI-410, TI-410	—	TI-402, PDI-406	FT-405	TI-403	—	—	AI-400	—
PFD stream	2	3	3	4	5	5	—	—	—	—	—	—	—

Valves: FCV-400 82.0 %, TCV-401 40.8 %, LCV-400 71.0 %, PCV-420 45.3 %. Analysers: AI-420 76.82 wt% (spec ≥ 75), AI-410 26.54 wt% (spec ≤ 28), TI-410 49.2 °C. Pressures: P-400 discharge 571.9, E-400 outlet 509.4, FCV-400 inlet 493.8 kPa g; hot-feed bubble pressure 359.1 kPa g (margin 134.7 kPa).

2 · Stream table, turndown (10 m³/h)

Stream	2	3	3A	4	5	5A	H1	H2	H3	H4	H5	N1	F1
Description	Feed to E-400 (P-400 discharge, battery limit)	Hot feed to FCV-400 (liquid under pressure)	Let-down outlet, flashing two-phase to D-400	Vapour to the fuel-gas header	D-400 liquid to LCV-400 (bubble point)	Bottoms after LCV-400 to the downstream unit (flashing)	Hot-oil supply to TCV-401 (header, total)	Hot oil to E-400 shell	Hot oil from E-400 shell	Hot-oil bypass (3-way valve)	Hot-oil return to header (mixed)	Nitrogen blanket (PCV-430)	PSV-400 relief to flare
From → to	P-400 (upstream unit) → E-400 tubes	E-400 tubes → FCV-400	FCV-400 / HV-401 → D-400 inlet N1	D-400 top (PCV-420) → Vapour header 30 kPa g	D-400 bottom N3 → LCV-400	LCV-400 / HV-411 → Downstream receiver 40 kPa g	Hot-oil supply header → TCV-401 (3-way)	TCV-401 → E-400 shell	E-400 shell → Return junction	TCV-401 → Return junction	Return junction → Hot-oil return header	N2 header 700 kPa g → D-400 top	D-400 (PSV-400) → Flare header
Phase	L	L	L+V	V	L	L+V	HO	HO	HO	HO	HO	N2	V
Vapour mass fraction	0.000	0.000	0.170	1.000	0.000	0.059	—	—	—	—	—	—	—
Temperature °C	10.0	74.0	49.3	49.2	49.2	40.2	180.0	180.0	111.7	180.0	168.6	28.0	49.2
Pressure kPa g	630.5	610.9	102.7	102.7	133.6	43.9	500.0	480.0	370.0	370.0	350.0	102.7	102.7
Mass flow kg/h	6,380	6,380	6,380	1,063	5,320	5,320	36,000	6,015	6,015	29,985	36,000	0	0
Molar flow kmol/h	86.54	86.54	86.54	16.91	69.66	69.66	—	—	—	—	—	0.00	0.00
Volume m³/h actual	10.0	11.2	235.3	222.1	8.8	97.8	45.0	7.5	7.5	37.5	45.0	0.0	—
Vapour Nm³/h	—	—	—	379	—	—	—	—	—	—	—	—	—
Density kg/m³	638.01	571.81	27.12	4.79	605.41	54.37	800.00	800.00	800.00	800.00	800.00	2.28	—
Butane wt%	35.00	35.00	35.00	76.82	26.63	26.63	—	—	—	—	—	—	—
Molar mass	73.72	73.72	73.72	62.86	76.36	76.36	—	—	—	—	—	28.01	62.86
Instruments	FT-400, TI-400	TT-401	(TI-410 in the drum)	FI-420, AI-420	FT-410, AI-410, TI-410	—	TI-402, PDI-406	FT-405	TI-403	—	—	AI-400	—
PFD stream	2	3	3	4	5	5	—	—	—	—	—	—	—

Valves: FCV-400 57.8 %, TCV-401 16.7 %, LCV-400 48.6 %, PCV-420 24.6 %. Analysers: AI-420 76.82 wt% (spec ≥ 75), AI-410 26.63 wt% (spec ≤ 28), TI-410 49.2 °C. Pressures: P-400 discharge 630.5, E-400 outlet 614.8, FCV-400 inlet 610.9 kPa g; hot-feed bubble pressure 359.1 kPa g (margin 251.9 kPa).

3 · Duties and closures

Quantity	Design	Maximum rate	Turndown
E-400 duty, feed side kW	420.2	525.3	262.6
E-400 duty, hot-oil side kW	420.2	525.3	262.6
D-400 heat loss kW	2.55	2.55	2.54
P-400 hydraulic kW (interface)	2.22	2.62	1.47
P-400 absorbed kW (η 55 %, assumed)	4.04	4.77	2.68
Mass in kg/h	10,208.2	12,760.2	6,380.1
Mass out kg/h	10,208.2	12,760.5	6,382.9
Mass closure %	0.0000	0.0024	0.0431
Butane in kg/h	3,572.9	4,466.1	2,233.0
Butane out kg/h	3,572.9	4,466.2	2,233.6
Butane closure %	0.0000	0.0020	0.0259

4 · Supply and let-down hydraulics

Model relation: feed m = √((P_sup0 − P_drum)/(k_sup + k_E-400 + k_line + 1/(ρ C²))) with C = C_FCV f(x) + C_HV-403 · opening (model step()); k_{sup} 6.2184, k_{E-400} 4.97472, k_{line} 1.24368 kPa/(kg/s)² (50 + 40 + 10 kPa at design), shut-off 650 kPa g, equal-% trim R = 30.

Valve	Case	Q m³/h	P1 → P2 kPa g	Pv kPa g	ΔP kPa	ΔPmax kPa	Choked	Cv √ΔP	Cv choked	Cv installed	Model OUT %	Opening choked %
FCV-400	Design	17.85	550.0 → 102.7	359.1	447.3	208.2	yes	7.38	10.82	18.17	73.5	84.7
FCV-400	Maximum rate	22.32	493.8 → 102.7	359.1	391.1	162.6	yes	9.86	15.30	18.17	82.0	94.9
FCV-400	Turndown	11.16	610.9 → 102.7	359.1	508.2	257.5	yes	4.33	6.08	18.17	57.8	67.8
LCV-400	Design	14.03	133.6 → 50.0	102.7	83.5	43.1	yes	13.81	19.22	47.98	63.4	73.1
LCV-400	Maximum rate	17.53	133.6 → 55.7	102.7	77.9	43.1	yes	17.87	24.01	47.98	71.0	79.6
LCV-400	Turndown	8.79	133.6 → 43.9	102.7	89.6	43.1	yes	8.35	12.03	47.98	48.6	59.3

IEC 60534-2-1: $\Delta P_{max} = F_L^2(P_1 - F_F P_v)$, $F_F = 0.96 - 0.28\sqrt{(P_v/P_c)}$, F_L 0.90 (globe, typical); P_c by Kay's rule. C_v (US) = 1316 × the model mass coefficient. Both valves flash at the outlet.

5 · Design conditions

System	Design P kPa g	Design T °C	Rule
Feed system to FCV-400 inlet	820.0	65.0	(1) on the P-400 shut-off 650 kPa g; (a)+(d)
Hot feed line	820.0	205.0	(1) on the P-400 shut-off; (b) hot-oil design temperature
D-400 system	450.0	150.0	(2) MAWP = PSV-400 set 450 kPa g ≥ rule (1) 280 kPa g; (c) 150 °C ≥ rule (a) 100 °C
Hot oil	1,000.0	205.0	(3) hot-oil system design; (b)
Nitrogen upstream of PCV-430	1,000.0	65.0	(3) nitrogen system design; (d)
Flare tail pipe	350.0	150.0	(3) flare tail pipe (assumed); relief temperature ≤ 150 °C

• (1) Design pressure = max(1.10 × maximum operating pressure, maximum operating pressure + 170 kPa), rounded up to 10 kPa. The maximum operating pressure is the highest of the three HMB cases; for everything that P-400 can pressurise (feed line, E-400 tubes, let-down valve inlet) it is the P-400 shut-off 650 kPa g.

• (2) The D-400 system (drum, let-down outlet, vapour and bottoms lines to their first spec break) takes the MAWP on the P&ID, 450 kPa g = PSV-400 set; it must not be lower than rule (1) and the PSV set must not exceed it (ASME VIII UG-134).

• (3) Utilities: hot-oil system 1000 kPa g, nitrogen upstream of PCV-430 1000 kPa g, flare tail pipe 350 kPa g (assumed system design pressures).

• (4) The wall thickness adds the static head of liquid with D-400 liquid-full to the top tangent (overfill) at the bottom of the drum (UG-22).

• (a) Design temperature = maximum operating temperature + 25 °C, rounded up to 5 °C.

• (b) Everything that the hot oil can heat with the feed stopped (both sides of E-400, the hot feed line to FCV-400) takes the hot-oil design temperature 180 + 25 = 205 °C.

• (c) The D-400 system takes the P&ID design temperature 150 °C (it covers rule (a) on the 74 °C inlet and the 102 °C hot feed with TCV-401 wide open).

• (d) Hydrocarbon equipment and lines in the sun: not below 65 °C (solar gain, assumed).

• (e) Minimum design metal temperature (MDMT) 0 °C: the cold feed (10 °C) less a 10 K margin. Depressuring cannot chill the metal further: the feed boils at 20.1 °C and the drum liquid at 26 °C at atmospheric pressure (model VLE).

6 · Engineering queries

No.	Subject	Query	Why it matters	Proposed answer	Status
EP-02	FCV-400 is in choked flashing service	At design FCV-400 takes ΔP 447.3 kPa while the choked limit is $\Delta P_{max} = F_L^2(P_1 - F_F P_v)$ = 208.2 kPa (F_L 0.90 typical). The model uses the square-root law over the whole ΔP.	Choked, the valve needs C_v 10.82 instead of 7.38 (opening 84.7 % instead of 73.5 % on the model's installed C_v 18.17); the trim must take flashing.	Specify FCV-400 for flashing service (angle body, hardened trim, outlet expansion) and size it on the choked ΔP (Instrumentation tab); keep the model as is and teach the difference.	OPEN
EP-04	LCV-400 outlet flashes	Across LCV-400 the liquid drops from 133.6 to 50 kPa g, below its bubble pressure 102.7 kPa g: 5.3 wt% flashes (stream 5A). LCV-400 is choked (ΔP 83.5 > ΔP_{max} 43.1 kPa). The model keeps the bottoms liquid down to the receiver.	Drawn 2" the outlet line would run at 17.36 m/s, ρv² 18927 Pa.	Number the outlet 4"-P-4005-A1A; specify LCV-400 for flashing; tell the downstream unit that its feed arrives two-phase.	OPEN

No.	Subject	Query	Why it matters	Proposed answer	Status
EP-06	FIC-400 high alarm differs	P&ID note 13 gives FIC-400 alarms 5 / 18 m³/h; the DCS alarm table (ALM) and the Design tab use 5 / 20 m³/h. The maximum-rate case (20 m³/h) sits on the DCS value.	Two documents disagree on an alarm; the operating window with alarms ends at the high alarm.	Confirm 20 m³/h (FIC-400 HI = maximum rate of the HMB) and revise note 13.	OPEN
EP-08	Hot-oil header pressures and properties assumed	The model has only the header ΔP (150 kPa) and c_p; supply / return pressures 500 / 350 kPa g, density and viscosity are assumed for the HMB and the line list.	The hot-oil line sizes, E-400 shell design pressure and XV-409 rating depend on them.	Obtain the hot-oil system data sheet (pressures, oil grade and properties) from the utility owner.	OPEN
EP-09	AI-420 sample at its dew point	The vapour leaves D-400 saturated at 49.2 °C; the sample line to AI-420 at ambient (28 °C) will condense and the GC will read the wrong composition.	Analyser reading drives the spec (AI-420 ≥ 75 wt%).	Heat-trace and insulate the AI-420 sample line to ≥ 60 °C (load HT-420 in EE-LL-001), or reduce the pressure at the tap.	OPEN